



The interface unit provides a place to mount the Pololu Motor Controller. It also provides a means to connect the system components including the Power Supplies, to Laser and moto controller; and connect the sensor signals to the analog inputs.

#### 2.1.3.1 Reference Current Source Circuit

The Current Source required to drive the sensor signal is implemented with one Linear Technologies LT3092 integrated circuit (IC) for each sensor channel. Parameters for consideration are the reference current ( $I_s$ ) as determined by the in situ current of the OLS to be tested; the compliance voltage ( $V_{dc}$ ) determined from the required maximum resistance to be measured at  $I_s$ . The Elevate OLS resistance ranges from  $100\Omega$  -  $1100\Omega$  with an overrange resistance of  $2100\Omega$  with an in situ of  $14.7\text{ma}$ , producing a maximum signal voltage  $V_s$  of  $16.7\text{V}$  at  $1100\Omega$ . We assume that the overrange resistance does not have to be accurately measured, only detected and that the Laser uses a  $24\text{V}_{DC}$  source that can be used to power the Current Source circuit  $V_{dc}$ , where the maximum resistance measurement will be  $1632\Omega$ . With  $V_{dc}$  set to nominal  $224\text{V}_{DC}$  following the LT3092 Datasheet Application Information page 9. Then  $15\Omega$  is selected for  $R_{OUT}$  and  $22.0\text{K}\Omega$  is calculated for  $R_{SET}$  as shown in Figure 2.

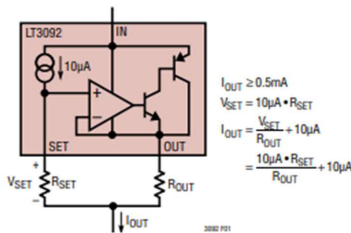


Figure 2. LT3092 Current Source Circuit

#### 2.1.3.2 Buffer/Scaler Circuit

The Buffer/scaler circuit addresses two issues with the DI-2108:

- Limited input voltage range
- Relatively low input impedance.

To achieve  $R_{in} > 1\text{M}\Omega$ , a LM2902 unity gain, follower circuit will buffer and isolate the DUT from the Analog Input impedance. With maximum signal voltage  $V_s > 16.7\text{V}$  and the maximum input voltage range of the DI-2108 of  $\pm 10.0\text{V}$ , then  $V_s$  must be scaled to fit within the input range. This is accomplished with voltage divider circuit consisting of  $R_7$  ( $R_9$ ) and  $R_8$  ( $R_{10}$ ). where the parallel resistance ( $R_{in} = R_7 \parallel R_9$ ) is greater than  $1\text{M}\Omega$ .

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To achieve  $R_{in} > 1M\Omega$ , the relatively low  $100K\Omega$  input resistance of the DI-2108 shown in Figure 3 needs to be increased by adding a single supply, follower op amp (LM2902).

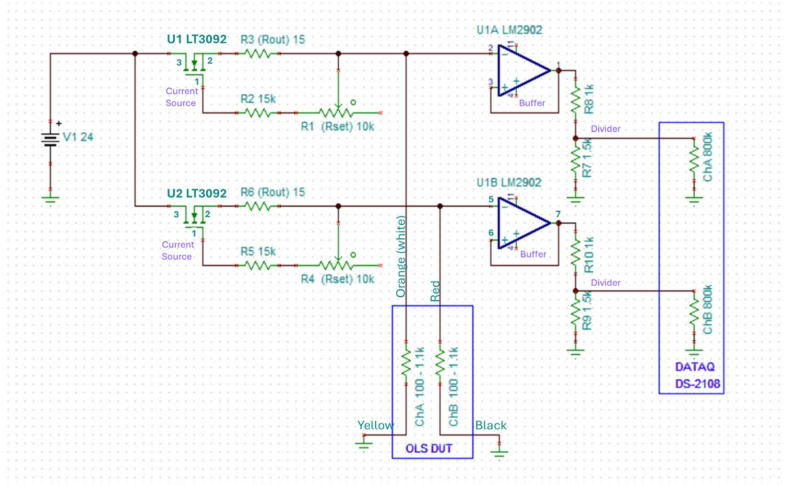


Figure 3. Analog Input Circuit

### Connectors

#### 2.1.4

##### 2.1.4.1 Breadboard implementation

##### 2.1.4.2 PCB implementation

### 3.0 Hardware Calibration

#### 3.1 Level Measurement Calibration

#### 3.2 Resistance Measurement Calibration

#### 3.3 Level Control Performance

Hardware status shall be indicated as Good=Green; Bad=Red; Needs attention/configuration or not automatic = Yellow i.e. calibration due (nice to have?)

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